

Summary of Increased Responsibilities

Moein Razavi, Data Scientist

August 2023 - Present

Role Expansion Overview

Initial Assignment (August 2023)

- Basic EV charging data analysis and processing tasks
- Supporting existing analytics workflows under direct supervision
- Standard reporting and documentation activities
- Learning Ford's proprietary systems and development tools

Current Expanded Responsibilities (Present)

Cloud Infrastructure & DevOps Leadership

- Design and deploy enterprise-scale cloud solutions using Google Cloud Run and Tekton CI/CD pipelines
- Lead serverless architecture migrations with automated deployment and scaling capabilities
- Manage production infrastructure serving critical Ford business applications

Artificial Intelligence & Machine Learning Innovation

- Architect intelligent agents for automated code complexity analysis, providing time/space optimization recommendations and GCP cost estimations via Ford's LLM API
- Develop state-of-the-art transformer-based forecasting models (TFT, Autoformer, Informer) using Hugging Face frameworks for EV predictive analytics
- Implement advanced hyperparameter optimization strategies using Optuna for enhanced model performance

Enterprise Data Engineering & Analytics

- Lead critical data processing pipelines for Ford's Energy Service Simulator using PySpark for large-scale data integration of simulator inputs and optimizer outputs
- Design and implement Astro Airflow orchestration architecture for automated data workflows
- Develop real-time EV discharge insights dashboards providing actionable business intelligence
- Build and maintain REST APIs for relevancy and operational excellence dashboards enabling seamless integration with GDI&A's Command Center systems